

LABORATORY RECORD

Course	Full Stack Web Development
Course Code	23CM4121
Experiment No.	3
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Date	29-08-2026

1. TITLE

Design a Responsive Webpage Using HTML5 and CSS3

2. AIM

To design and develop a modern, responsive webpage named Creative Web using HTML5 and CSS3, consisting of a header with navigation, a home section, a services section with service cards, a contact section, and a footer.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 webpage.
- To design a webpage using HTML5 semantic elements.
- To apply CSS3 styling to HTML elements using an external stylesheet.
- To create a navigation bar using CSS.
- To design a home/welcome section with a call-to-action button.
- To create service cards using CSS Flexbox.
- To apply hover effects on navigation links, buttons, and cards.
- To make the webpage responsive using CSS media queries.
- To understand the separation of HTML structure and CSS presentation.

4. THEORY

HTML5 (HyperText Markup Language 5) is used to define the structure and content of a webpage. It provides elements such as <header>, <nav>, <section>, and <footer> for organizing webpage content.

CSS3 (Cascading Style Sheets 3) is used to control the presentation and appearance of HTML elements. CSS can be used to define colors, fonts, spacing, layouts, borders, shadows, hover effects, and responsive behavior.

In this experiment, HTML5 is used to create the structure of a website called Creative Web, while an external CSS file named style.css is used for styling.

The webpage consists of:

- Header: Contains the website title (Creative Web) and navigation links.
- Home Section: Contains a welcome heading, a short description, and a "Get Started" button.
- Services Section: Displays three service cards (Web Design, Development, Support) arranged using Flexbox.
- Contact Section: Contains a heading, a short description, and a "Contact Us" button linked to an email address using mailto.
- Footer: Displays copyright information.

CSS3 features such as Flexbox, border-radius, box-shadow, hover effects, and media queries are used to improve the appearance and responsiveness of the webpage.

The basic working flow is:

HTML Structure → CSS Styling → Browser Rendering → Responsive Webpage

5. ALGORITHM / PROCEDURE

1. Create a new project folder for the webpage.
2. Create an HTML file named static_webpage.html.
3. Create an external CSS file named style.css.
4. Define the HTML5 document structure using <!DOCTYPE html>.
5. Add the webpage title and link the external CSS file.
6. Create a header containing the website title and navigation links.
7. Create a home section with a heading, description, and a "Get Started" button.
8. Create a services section containing three service cards using <div> elements.
9. Add a heading, description, and "Learn More" link inside each service card.
10. Create a contact section with a heading, description, and a mailto contact button.
11. Create a footer containing copyright information.
12. Use CSS to reset default margins and padding, and enable smooth scrolling.
13. Apply Flexbox to arrange the service cards in the services section.
14. Apply colors, spacing, typography, border-radius, and box-shadow using CSS.
15. Add hover effects to navigation links and buttons.
16. Implement responsive design using the @media rule for smaller screens.
17. Save both HTML and CSS files in the same project folder.
18. Open static_webpage.html in a web browser and verify the complete webpage output.

6. PROGRAM

static_webpage.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Creative Web</title>

  <!-- CSS File -->
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Header -->
  <header>
    <h1>Creative Web</h1>

    <nav>
      <a href="#home">Home</a>
      <a href="#services">Services</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <!-- Home Section -->
  <section class="home" id="home">

    <h2>Welcome to My Website</h2>

    <p>
      We create simple and beautiful websites.
    </p>

    <a href="#services" class="btn">Get Started</a>

  </section>

  <!-- Services Section -->
  <section class="services" id="services">

    <h2>Our Services</h2>

    <div class="cards">

      <div class="card">
        <h3>Web Design</h3>

        <p>
          We create attractive and responsive website designs
          for different types of businesses.
        </p>

        <a href="#contact" class="card-btn">Learn More</a>
      </div>
```

```

        <div class="card">
            <h3>Development</h3>

            <p>
                We develop fast, simple and user-friendly websites
                using modern web technologies.
            </p>

            <a href="#contact" class="card-btn">Learn More</a>
        </div>

        <div class="card">
            <h3>Support</h3>

            <p>
                We provide website support and maintenance to keep
                your website running smoothly.
            </p>

            <a href="#contact" class="card-btn">Learn More</a>
        </div>

    </div>

</section>

```

```

<section class="contact" id="contact">

    <h2>Contact Us</h2>

    <p>
        Have a project in mind? Get in touch with us today.
    </p>

    <a href="mailto:example@gmail.com" class="btn">
        Contact Us
    </a>

</section>

<!-- Footer -->
<footer>

    <p>
        &copy; 2026 Creative Web. All Rights Reserved.
    </p>

</footer>

</body>

</html>

```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

html {
  scroll-behavior: smooth;
}

body {
  font-family: Arial, sans-serif;
  background: #f4f4f4;
  color: #222;
}

/* Header */
header {
  background: #2563eb;
  color: white;
  text-align: center;
  padding: 30px;
}

header h1 {
  margin-bottom: 15px;
}
```

```
nav a {
  color: white;
  text-decoration: none;
  margin: 0 12px;
}

nav a:hover {
  color: red;
}

/* Home */
.home,
.contact {
  background: white;
  text-align: center;
  padding: 60px 20px;
}

.home h2,
.contact h2 {
  color: #2563eb;
  margin-bottom: 15px;
}

.home p,
.contact p {
  margin-bottom: 25px;
}
```

```

.btn,
.card-btn {
  background: #2563eb;
  color: white;
  padding: 10px 20px;
  text-decoration: none;
  border-radius: 5px;
  display: inline-block;
}

.btn:hover,
.card-btn:hover {
  background: palevioletred;
}

/* Services */
.services {
  padding: 50px 20px;
  text-align: center;
}

.services h2 {
  margin-bottom: 30px;
}

.cards {
  display: flex;
  gap: 20px;
  justify-content: center;

```

```

.card {
  background: white;
  padding: 25px;
  width: 30%;
  border-radius: 8px;
  box-shadow: 0 3px 8px #ccc;
}

.card h3 {
  color: #2563eb;
  margin-bottom: 12px;
}

.card p {
  margin-bottom: 20px;
}

/* Footer */
footer {
  background: #222;
  color: white;
  text-align: center;
  padding: 20px;
}

/* Mobile */
@media (max-width: 700px) {
  nav a {
    display: block;

```

```

        margin: 8px;
    }

    .cards {
        flex-direction: column;
        align-items: center;
    }

    .card {
        width: 90%;
    }
}

```

7. CODE EXPLANATION

Code	Explanation
<!DOCTYPE html>	Declares the document as an HTML5 document.
<meta charset="UTF-8">	Defines the character encoding as UTF-8.
<meta name="viewport"...>	Makes the webpage responsive on different screen sizes.
<title>	Specifies the title displayed in the browser tab.
<link rel="stylesheet"...>	Connects the external CSS file with the HTML document.
<header>	Defines the header section containing the title and navigation.
<h1>	Displays the main website title, "Creative Web".
<nav>	Contains the main navigation links.
<section>	Defines individual sections of the webpage (home, services, contact).
id	Provides a unique identifier for internal navigation.
<div class="cards">	Groups the three service cards using Flexbox.
<div class="card">	Represents an individual service card.
	Creates styled button-like links.
href="mailto:..."	Opens the user's default email client to send a message.
<footer>	Defines the footer section.
* { margin, padding, box-sizing }	Resets default spacing and sets consistent box sizing.
scroll-behavior: smooth	Enables smooth scrolling between sections.
display: flex	Arranges the service cards in a flexible row layout.

Code	Explanation
<code>gap</code>	Adds spacing between flex items.
<code>justify-content</code>	Aligns flex items horizontally.
<code>border-radius</code>	Creates rounded corners for buttons and cards.
<code>box-shadow</code>	Adds a shadow effect to the service cards.
<code>:hover</code>	Changes link, button, and card colors when the mouse pointer moves over them.
<code>@media (max-width: 700px)</code>	Defines responsive styles for smaller screens.
<code>flex-direction: column</code>	Stacks the service cards vertically on small screens.

8. TEST CASES

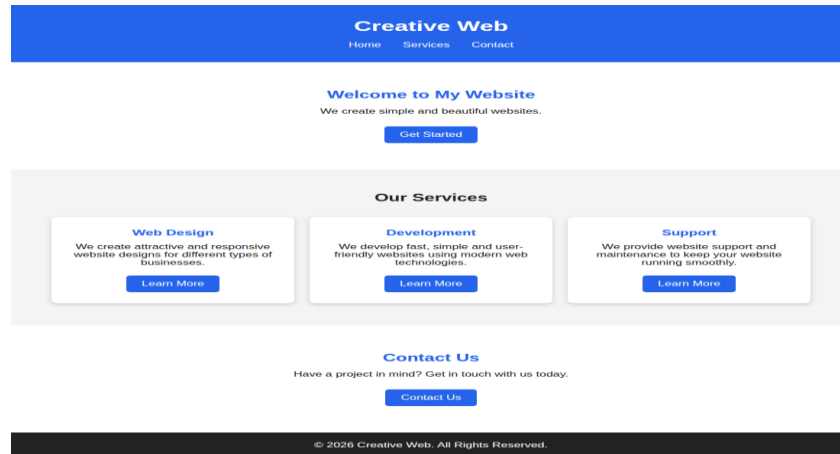
The webpage was executed in a web browser and tested for navigation, layout, styling, hover effects, the contact link, and responsive behavior.

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Open static_webpage.html	Complete webpage should be displayed	Webpage displayed successfully	Pass
TC-02	Click Home	Page should navigate to Home section	Home section displayed	Pass
TC-03	Click Services	Page should navigate to Services section	Services section displayed	Pass
TC-04	Click Contact	Page should navigate to Contact section	Contact section displayed	Pass
TC-05	Click Get Started button	Page should navigate to Services section	Services section displayed	Pass
TC-06	Hover over navigation links	Link colour should change to red	Hover effect displayed	Pass
TC-07	Hover over Get Started / Contact Us button	Button background should change to palevioletred	Hover effect displayed	Pass
TC-08	Hover over a service card's Learn More button	Button background should change to palevioletred	Hover effect displayed	Pass
TC-09	Click Contact Us button	Default email client should open with the recipient address filled	Email client opened successfully	Pass
TC-10	Resize browser to mobile width	Service cards and nav links should stack vertically	Responsive layout displayed	Pass

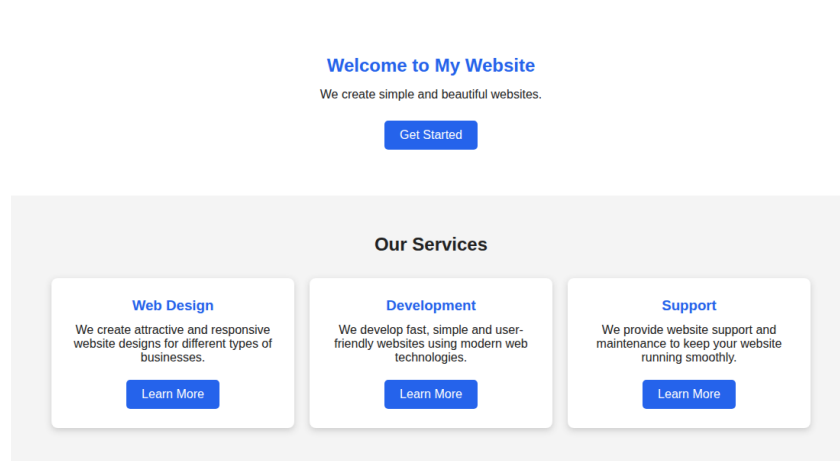
9. OUTPUT

Each test case was executed in the browser individually and the rendered/observed output was captured below, corresponding to the test cases in Section 8.

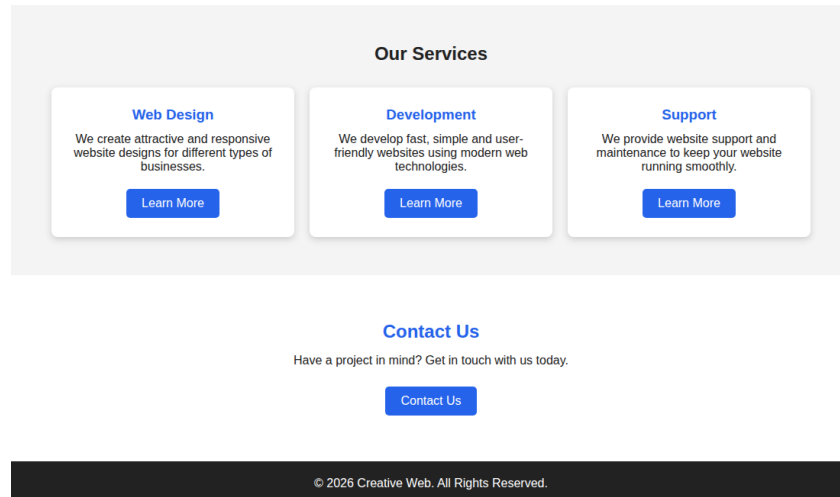
Output 1 (TC-01): Open static_webpage.html



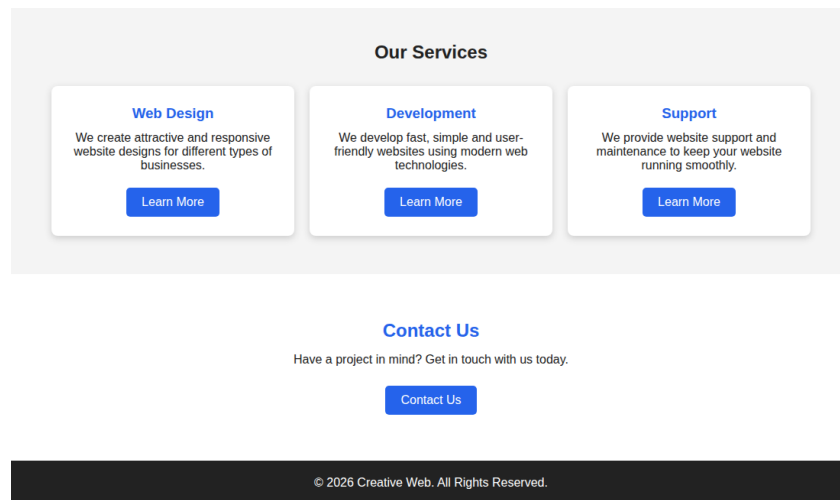
Output 2 (TC-02): Click Home



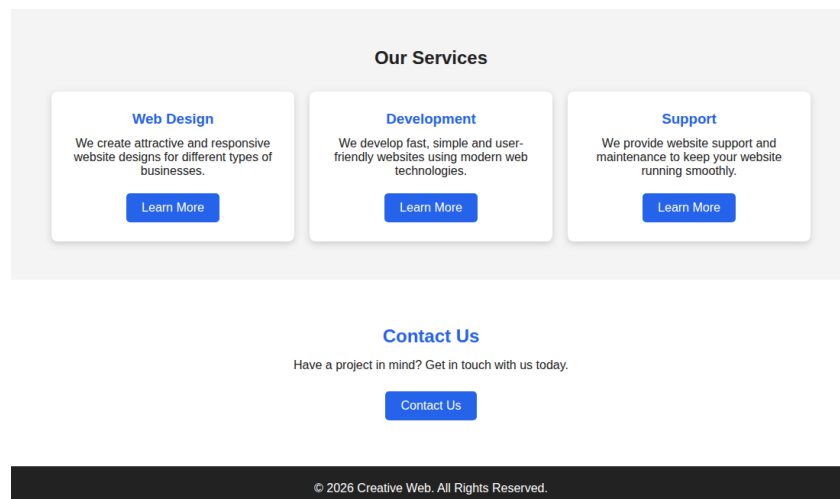
Output 3 (TC-03): Click Services



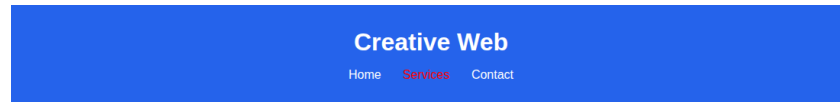
Output 4 (TC-04): Click Contact



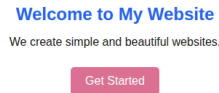
Output 5 (TC-05): Click Get Started button



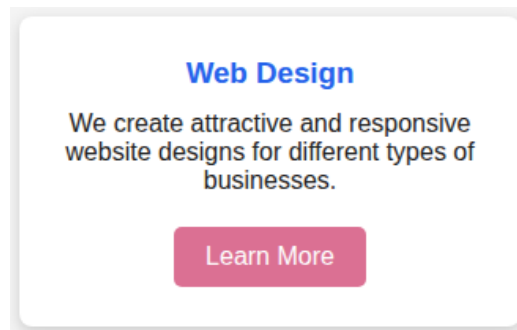
Output 6 (TC-06): Hover over navigation links



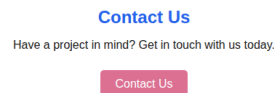
Output 7 (TC-07): Hover over Get Started / Contact Us button



Output 8 (TC-08): Hover over a service card's Learn More button



Output 9 (TC-09): Click Contact Us button



Output 10 (TC-10): Resize browser to mobile width

Creative Web

[Home](#)
[Services](#)
[Contact](#)

Welcome to My Website

We create simple and beautiful websites.

[Get Started](#)

Our Services

Web Design

We create attractive and responsive website designs for different types of businesses.

[Learn More](#)

Development

We develop fast, simple and user-friendly websites using modern web technologies.

[Learn More](#)

Support

We provide website support and maintenance to keep your website running smoothly.

[Learn More](#)

Contact Us

Have a project in mind? Get in touch with us today.

10. RESULT

Thus, a modern and responsive webpage named Creative Web was successfully designed and developed using HTML5 and CSS3. The webpage contains a navigation bar, home section, services section with service cards, a contact section, and a footer. CSS3 features such as Flexbox, border-radius, box-shadow, hover effects, and media queries were successfully implemented. The webpage was executed and verified successfully in a web browser, with individual outputs captured for each test case above.